

Week 2 Technical Report

Neural Networks and CNN Image Classification

1. Introduction

In Week 2 of my internship, I learned about neural networks and image classification. I implemented a Feed-Forward Neural Network and a Convolutional Neural Network (CNN) using TensorFlow and Keras in Google Colab.

2. Dataset

For the CNN project, I used the **CIFAR-10 dataset**. It contains 50,000 training images and 10,000 test images. The images are RGB images of size 32×32 pixels and belong to 10 different classes such as airplane, automobile, bird, cat, dog, frog, horse, ship, and truck.

3. Feed-Forward Neural Network

First, I implemented a simple Feed-Forward Neural Network to understand the basic working of neural networks. The model contained input, hidden, and output layers. ReLU was used in the hidden layers and Sigmoid was used in the output layer. The model was trained using the Adam optimizer.

4. CNN Image Classifier

A CNN was developed for CIFAR-10 image classification. The model included convolutional layers, max-pooling layers, a flatten layer, and dense layers. The images were normalized by scaling pixel values from 0–255 to 0–1.

The CNN was trained using the Adam optimizer and evaluated using test accuracy and loss.

5. Hyperparameter Experiments

Different hyperparameters were tested to understand their effect on model performance.

Experiment	Epochs	Batch Size	Learning Rate	Test Accuracy
Baseline	10	64	Default	67.01%
More Epochs	20	64	Default	67.95%
Lower Learning Rate	10	64	0.0001	53.81%
Smaller Batch Size	10	32	Default	65.73%

6. Results

The baseline CNN achieved **67.01% test accuracy**. Increasing the training epochs to 20 resulted in **67.95% test accuracy**. Using a learning rate of 0.0001 resulted in **53.81% accuracy**, while reducing the batch size to 32 resulted in **65.73% accuracy**.

These experiments showed that changing hyperparameters can affect the learning and performance of a CNN.

7. Conclusion

During Week 2, I gained practical experience with Feed-Forward Neural Networks and CNNs. I learned how to train an image classification model, evaluate its performance, and experiment with different hyperparameters. This helped me understand the basic workflow of deep learning and CNN-based image classification.